

PMS Intelligence and Overdue Work Risk Analysis

DOCUMENT

Management Summary

PROGRAM

AI Residency Grouping Challenge — July 2026

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Completion Rate	On-Time Rate	Calc. Overdue Rate	Stored-Field Error	Model ROC-AUC
94.3%	50.6%	51.6%	35.3% of tasks	0.676

Overall PMS Data Reliability

PMS data can support reliable reporting and prediction, but only after correction: the stored **is_overdue** field understates true overdue risk by 3x (17.4% stored vs. 51.6% calculated), and 8 core tables carry exact-duplicate rows that silently fan out task counts if joined without deduplication.

Most Important Data-Quality Problems

- Stored overdue flag disagrees with reality on 35.3% of tasks
- 89.3% of tasks have no direct department_id (position used instead)
- 8 core tables duplicated 2:1; all joins require DISTINCT ON (id)
- 664 tasks (4.8%) have no assigned position
- 138 completed tasks still have open sub-tasks

Departments Requiring Attention

Overdue rates range 11.1%–82.8% by department. **EDGE Program** is the top priority: high overdue rate combined with high volume (1,808 tasks). Small-sample departments (e.g. Finance, 29 tasks) show extreme rates that should be read with caution.

Main Factors Associated with Overdue Work

- Whether a task has started by day 7 is the single dominant risk signal (95.7% of model importance)
- Planned tasks run overdue more than unplanned tasks (56.8% vs. 29.6%) — counterintuitive, needs review
- Task weight shows only weak correlation with delay ($r = 0.14$)

Model Performance

GradientBoostingClassifier, time-based split (1,213 train / 315 test). ROC-AUC 0.676, recall 1.0 on the test set. In practice the model behaves as a single early-warning rule (“has this task started by day 7?”) rather than a nuanced multi-factor risk score.

Major Model Limitations

- One feature (started_late_cat) drives 95.7% of predictions — low nuance
- Zero false negatives reflects extreme feature separation, not proven generalization
- 44 false positives cluster in zero-weight, “unknown-start” tasks — likely missing data, not true risk

Recommended PMS Data Improvements

- Fix is_overdue to update automatically once a deadline passes
- Enforce task.department_id population, or deprecate it in favor of position-derived department
- Log actual_start_date reliably to remove the “unknown-start” ambiguity driving most model error
- Prevent duplicate-row insertion at the source rather than relying on downstream DISTINCT logic

Recommended Use of the Prediction Model

Use as an early-warning trigger for planning and resource conversations only — e.g. flagging tasks that have not started 7 days after their planned start for a manager check-in.

Governance: PMS analytics and machine learning predictions should support early intervention, planning, and resource allocation. They must not be used as the sole basis for employee ranking, disciplinary action, promotion, termination, or compensation decisions.